Week 12 – Advanced Social Media Platform

DG Interns Hub | Project Documentation

Project overview

Build a learning-focused social platform prototype demonstrating real-time chat, notifications, optional media upload, API testing, security basics, CI and career preparation.

Architecture

React browser client communicates with Express using REST and Socket.IO. Socket.IO rooms route chat events and personal notifications. Multer handles uploads and Cloudinary stores media when configured. Database persistence is a future extension.

Real-time features

The server demonstrates joining user/chat rooms, sending messages, new-message events, notifications and typing indicators. Authenticate sockets and verify room membership before production use.

File upload

Multer uses memory storage, a 10 MB size limit and an allowlist for JPEG, PNG, WebP and MP4. Cloudinary credentials are optional environment variables. Add stronger content inspection and quotas for production.

Testing strategy

Jest and Supertest examples validate the health endpoint and caption validation. Manually test two-tab chat, disconnect behavior, file upload, invalid formats and oversized files. Save genuine screenshots and test logs.

Security

Helmet, CORS configuration, API rate limiting, JSON size limit and basic validation are included. Full authentication, authorization, CSRF strategy where relevant, audit logging and database protections are not included.

Performance and SEO

The frontend is responsive and has basic metadata. Further improvements: lazy loading, code splitting, image optimization, caching, compression, accessibility checks, sitemap/robots and Lighthouse measurement. No performance score is claimed.

Deployment and DevOps

Docker backend example and GitHub Actions test workflow are included. CI runs tests only; automatic production deployment is not configured. Add hosting-specific steps and secrets after selecting providers.

Setup

Backend: copy .env.example to .env, run npm install and npm run dev. Frontend: npm install and npm run dev. Tests: cd backend && npm test.

Submission structure

Includes frontend/, backend/, tests/, docker/, .github/workflows/, portfolio-website/, resume/, docs/, README.md and architecture-diagram.svg. Export SVG to PNG only if the submission portal specifically requires PNG.

Career preparation

Be ready to explain REST vs WebSockets, Socket.IO rooms, file validation, rate limiting, automated tests, CI and environment variables. Add only verified project work to your resume.

Known limitations

No database persistence, complete authentication, full social feed, moderation, end-to-end browser tests, verified live deployment or measured Lighthouse results. This is a learning starter, not a complete production platform.

Quick commands

```
cd backend\ncp .env.example .env\nnpm install\nnpm run dev\n\n# second terminal\ncd frontend\nnpm install
```